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THE RVfpga MSc PROJECT

Porting RVfpgaEL2 to a New FPGA Board

0. INTRODUCTION

RVfpga is a general-purpose System-on-Chip (SoC) design that can be implemented on a wide range of FPGA development boards. When adapting the design to a new board, the main tasks involve modifying the internal peripheral controllers, redefining the external interface connections, and updating the configuration files that handle communication with the target FPGA board.

This document describes the complete process of porting **RVfpgaEL2** from the **Nexys A7** version to the **Nexys Video** board. The migration is implemented using **Vivado** for synthesis and hardware implementation, and the **Catapult SDK** for uploading and debugging. This serves as a reference workflow for adapting **RVfpgaEL2** to other FPGA platforms. The main steps are as follows:

1. Design and Constrain Modifications
2. Design Synthesis
3. Update the **OpenOCD** Configuration Files
4. Simulation and Verification

Some configuration files used in this guide are derived from existing open-source libraries and resources. For certain FPGA boards, additional configuration files may need to be created manually, and such board-specific adaptations are beyond the scope of this document.

1. DESIGN AND CONSTRAINTS MODIFICATIONS

Design

The target FPGA board may contain different peripherals or external memory devices compared to the original hardware. In many cases, existing peripheral controllers are not directly compatible, so updates must be made according to both the design requirements and the specific hardware features of the target device.

In this example, the Nexys Video board does not include an accelerometer; it only provides eight LEDs and switches, and requires adaptation to support the onboard OLED display. Within the `rvfpganexys` module, the signals `i_sw_slice[15:8]` and `gpio_out[15:8]` are not connected to any external input or output. During synthesis, these unconnected and nonessential paths will be automatically optimized out.

In the `veerwolf_core`, the original `simple_spi` instance (`spi2`), which was used for accelerometer communication, is replaced with the `oled_spi` instance (`spi2`). The `wb_intercon` module is regenerated to establish the correct connections and address mapping. Detailed information about these changes can be found in **Lab06** (`wb_intercon`) and **Lab10** (`oled_spi`).

Constraints

Constraints should match the top-level interfaces. Standard templates for Digilent FPGA boards can be found at [Digilent/digilent-xdc: A collection of Master XDC files for Digilent FPGA and Zynq boards](#). The signal names in `RVfpga` differ from the official templates, so modify the constraints file accordingly.

2. SYNTHESIS

Refer to **Lab05** for more detailed information on the synthesis process.

3. UPDATE THE OPENOCD CONFIGURATION FILES

This section outlines the configuration files that you may need to update to support FPGA programming and debugging. Four key files are involved in the porting process:

The screenshot shows the 'Configure Platforms' window in the Catapult-SDK. The platform selected is 'RVfpga Nexys EL2 (No DDR) via OpenOCD'. The interface includes several sections: 'Platform Description' (RVfpga is a recommended hardware configuration for the Imagination University Program...), 'Platform Name' (RVfpga Nexys EL2 (No DDR) via OpenOCD), 'Build configuration' (RVFPGANEXYS_EL2_NODDR), 'Connection Method' (OpenOCD (hardware)), 'Auto-run commands' (set pagination off; set confirm off; monitor reset init; monitor riscv semihosting enable; load [gdb commands to be run on connection to target (optional)]), 'Trace encoder configuration', 'Core Description File' (with a red box around it and a 'Choose file...' button), 'SVD File' (with a red box around it and a 'Choose file...' button), 'OpenOCD Config File' (with a red box around it and a 'Choose file...' button), 'FPGA bitfile' (with a red box around it and a 'Choose file...' button), and 'OpenOCD configuration file for FPGA programming' (with a red box around it and a 'Choose file...' button). At the bottom, there is a 'Program FPGA' button.

Figure 1. Catapult-SDK Configure Platforms

1. Core Description File - RVfpgaNexysEL2_NoDDR.YAML

This YAML file defines the starting address and size of the main memory regions used by RVfpgaEL2. You only need to modify this file when making changes to the el2_mem module or when using a different DDR memory configuration. You may find it at:

`[catapult-sdk]/build/platform/RVfpgaRVfpgaNexysEL2_NoDDR.YAML`

2. SVD File - RVfpgaNexys.svd

The .svd file provides information for the debugger to display user interface labels and details for memory-mapped addresses during debugging. Modifying this file

allows the debugger to display the corresponding address and register information in the UI. However, the file appears to be auto-generated by an unavailable script, so manual edits must follow the structure of existing peripherals. In practice, these changes have limited effect, and modification is generally not recommended.

If you wish to see the corresponding memory address hints during debugging, adjust the file path as follows:

[catapult-sdk]/build/platform/RVfpga/RVfpgaNexys.svd

3. OpenOCD Config File - RVfpgaNexysVideoEL2_NoDDR_debug.cfg

This file defines communication and memory mapping for the debugging process. The communication configuration is identical to the **OpenOCD Configuration File for FPGA Programming**, but includes additional sections that describe the address map used by the design. This enables access to specific memory addresses during debugging. If new peripherals are added that use different address mappings, the corresponding configuration entries must be updated.

The file used here is a modified version, with only the communication section changed. The modified file is located at:

[RVfpga_NexysVideo-NoDDR]/NexysVideo_config/RVfpgaNexysVideoEL2_NoDDR_debug.cfg

In Figure 2. Lines 19 to 20 define the main memory address range, while lines 23 to 28 specify the peripheral mapping addresses, which should correspond to the configuration used in:

[RVfpga_NexysVideo-NoDDR]\src\VeeRwolf\Interconnect\WishboneInterconnect

```
18 # Automatically imported from RVfpgaNexysEL2_NoDDR.YAML
19 ram bank 0xf0040000 0x00004000 $_TARGETNAME
20 ram bank 0x00000000 0x00010000 $_TARGETNAME
21
22 # Automatically imported from RVfpgaNexys.svd
23 ram bank 0x80001000 0x00000040 $_TARGETNAME # SWERVOLF_SYSTEM_CONTROLLER
24 ram bank 0x80001040 0x00000040 $_TARGETNAME # SWERVOLF_SPI_1
25 ram bank 0x80001100 0x00000040 $_TARGETNAME # SWERVOLF_SPI_2
26 ram bank 0x80001200 0x00000040 $_TARGETNAME # SWERVOLF_PTC
27 ram bank 0x80001400 0x00000040 $_TARGETNAME # SWERVOLF_GPIO
28 ram bank 0x80002000 0x00001000 $_TARGETNAME # SWERVOLF_UART
```

Figure 2. RVfpgaNexysVideoEL2_NoDDR_debug.cfg

4. FPGA Bitfile - rvfpganexys.bit

Fill in your bitstream file path. The bitfile location and generation process are described in **Lab05**.

5. OpenOCD Configuration File for FPGA Programming - RVfpgaNexysVideoEL2_NoDDR.cfg

RVfpgaEL2 uses the Catapult SDK for programming and debugging, which relies on OpenOCD for these operations. When programming the FPGA, the configuration must include definitions for FTDI adapter setup, target device setup, bitstream handling, and execution flow, as illustrated in the following figure. You can find this file at:

`[RVfpga_NexysVideo-NoDDR]\NexysVideo_config\RVfpgaNexysVideoEL2_NoDDR.cfg`

6. Other Configurations

If you plan to make major modifications, for example, extensive changes to the VeeRwolf core or adjustments to the system clock frequency, and need to distinguish these updates from configurations used by other boards, you may need to modify

`[RVfpga_NexysVideo-NoDDR]/common/Common.cmake`

and other files within the common directory.

These modifications are not required for this project and are therefore not listed here.

Using existing and verified configuration files is often more efficient and reliable. The communication section used in this example comes from [riscv-openocd/tcl/board at riscv · riscv-collab/riscv-openocd](#), and additional configuration files for other board are also provided by the VeeRwolf project at [VeeRwolf/data at main · chipsalliance/VeeRwolf](#).

4. SIMULATION AND VERIFICATION

If any design files were added or removed during the previous steps, make sure to update the simulation file list accordingly.

The relevant configuration file is located at:

`RVfpga/RVfpgaEL2/RVfpga_NexysVideo-NoDDR/Simulators/verilatorSIM_Trace/rvfpgasim.vc`

You may need to modify the Verilog include paths and file references in this file.

For example, if a new peripheral such as the OLED SPI module was added, include its path as shown below:

```
18 +incdir+../../src/VeeRwolf/Peripherals/spi_oled
19 -CFLAGS -I../../src/VeeRwolf/Peripherals/spi_oled
```

Add or remove file entries as needed to match the updated project structure, include new files path as shown below:

```
151 ../../src/VeeRwolf/Peripherals/spi_oled/oled_spi.v
```